

Practice Problems & Solutions

First-Order Ordinary Differential Equations

This workbook contains 25 practice problems covering all five techniques taught in the slide deck: Separation of Variables, Linear ODEs (Integrating Factor), Exact Equations, Non-Exact Equations with Integrating Factors, and Numerical Methods (Euler, Heun, RK4). Each section contains five problems of increasing difficulty, from basic technique drills to applied real-world scenarios. Full worked solutions follow each problem, using the same color-coding as the slides: **purple** for live mathematical steps, **orange** for final answers, and **blue** for structural labels.

Section	Topic	Problems	Key Technique
1	Separation of Variables	1.1 – 1.5	Separate $dy/dx = g(x)h(y)$, integrate both sides
2	Linear First-Order ODEs	2.1 – 2.5	Standard form $y' + P(x)y = Q(x)$, use $\mu = e^{\int P dx}$
3	Exact Equations	3.1 – 3.5	Test $M_y = N_x$, find potential $F(x,y) = C$
4	Non-Exact + Integrating Factor	4.1 – 4.5	Find $\mu(x)$ or $\mu(y)$, then solve as exact
5	Numerical Methods	5.1 – 5.5	Euler, Heun, RK4 — step along slope field

SECTION 1

Separation of Variables

Solve by separating $dy/dx = g(x)h(y)$ into $(1/h(y)) dy = g(x) dx$, then integrating both sides. Always check for equilibrium solutions where $h(y) = 0$.

Problem 1.1 — Basic Separation

PROBLEM:

Solve the differential equation:

$$dy/dx = x^2 y^2$$

SOLUTION:

Step 1 — Separate variables:

$$dy/y^2 = x^2 dx$$

Step 2 — Integrate both sides:

$$\int y^{-2} dy = \int x^2 dx \rightarrow -1/y = x^3/3 + C$$

Step 3 — Solve for y:

$$y = -1 / (x^3/3 + C) = -3 / (x^3 + 3C)$$

Also note $y = 0$ is an equilibrium solution (check: $h(y) = y^2 = 0$ when $y = 0$).

$$y = -3/(x^3 + C) \text{ or } y = 0$$

Problem 1.2 — IVP with Initial Condition

PROBLEM:

Solve $dy/dx = y \cos(x)$ with $y(0) = 1$.

SOLUTION:

$$\text{Step 1 — Separate: } dy/y = \cos(x) dx$$

$$\text{Step 2 — Integrate: } \ln|y| = \sin(x) + C$$

$$\text{Step 3 — Exponentiate: } y = Ae^{\sin(x)} \text{ (where } A = \pm e^C)$$

$$\text{Step 4 — Apply } y(0) = 1: 1 = Ae^0 \rightarrow A = 1$$

$$y(x) = e^{\sin(x)}$$

Problem 1.3 — Newton's Law of Cooling

PROBLEM:

A body cools from 80°C to 60°C in 5 minutes in a room at 20°C . Find $T(t)$ and predict $T(10)$.

SOLUTION:

Step 1 – Model: $dT/dt = -k(T - 20)$, $T(0) = 80$, $T(5) = 60$

Step 2 – Substitute $u = T - 20$: $du/dt = -ku \rightarrow du/u = -k dt$

Step 3 – Integrate: $\ln|u| = -kt + C \rightarrow u = Ae^{-kt} \rightarrow T = 20 + Ae^{-kt}$

Step 4 – Apply $T(0) = 80$: $80 = 20 + A \rightarrow A = 60$

Step 5 – Apply $T(5) = 60$: $60 = 20 + 60e^{-5k} \rightarrow e^{-5k} = 40/60 = 2/3 \rightarrow k = (1/5)\ln(3/2) \approx 0.0811$

Step 6 – Predict: $T(10) = 20 + 60e^{-0.811} \approx 20 + 60(0.4444) \approx 46.7^\circ\text{C}$

$$T(t) = 20 + 60e^{-0.0811t}, \quad T(10) \approx 46.7^\circ\text{C}$$

Problem 1.4 — Logistic Growth

PROBLEM:

Solve the logistic equation $dP/dt = rP(1 - P/K)$ with $P(0) = P_0$. (Let r and K be positive constants.)

SOLUTION:

Step 1 – Partial fractions: $dP / [P(1 - P/K)] = r dt$

Step 2 – Decompose: $1/[P(1-P/K)] = K/[P(K-P)] = 1/P + 1/(K-P)$

Step 3 – Integrate: $\int (1/P + 1/(K-P)) dP = \int r dt \rightarrow \ln|P| - \ln|K-P| = rt + C$

Step 4 – Exponentiate: $P/(K-P) = Ae^{rt}$, where $A = P_0/(K-P_0)$

Step 5 – Solve for P : $P = K Ae^{rt} / (1 + Ae^{rt}) = KP_0 e^{rt} / (K - P_0 + P_0 e^{rt})$

$$P(t) = KP_0 e^{rt} / (K - P_0 + P_0 e^{rt})$$

Problem 1.5 — Torricelli's Law (Draining Tank)

PROBLEM:

A cylindrical tank of radius $R = 2$ m drains through a hole of area $a = 0.01$ m². Torricelli's law gives $dV/dt = -a\sqrt{2gh}$, where $V = \pi R^2 h$. Find $h(t)$ and the time to drain from $h_0 = 4$ m.

SOLUTION:

Step 1 – Substitute $V = \pi R^2 h$: $\pi R^2 dh/dt = -a\sqrt{2gh}$

Step 2 – Separate: $dh/\sqrt{h} = -(a\sqrt{2g})/(\pi R^2) dt$

Step 3 – Integrate: $2\sqrt{h} = -(a\sqrt{2g})/(\pi R^2) t + C$. Apply $h(0) = h_0$: $C = 2\sqrt{h_0}$

Step 4 – Solve: $\sqrt{h} = \sqrt{h_0} - (a\sqrt{2g})/(2\pi R^2) t$

Step 5 – Time to drain ($h = 0$): $t = (2\pi R^2 \sqrt{h_0}) / (a\sqrt{2g})$

Step 6 – Plug in $R=2$, $a=0.01$, $h_0=4$, $g=9.8$: $t = (2\pi \cdot 4 \cdot 2)/(0.01 \cdot \sqrt{19.6}) \approx 50.24/0.04427 \approx 1135$ s ≈ 18.9 min

$$h(t) = (\sqrt{h_0} - (a\sqrt{2g})/(2\pi R^2) \cdot t)^2, \quad t_{\text{drain}} \approx 18.9 \text{ min}$$

SECTION 2

Linear First-Order ODEs

Put the ODE in standard form $y' + P(x)y = Q(x)$, compute the integrating factor $\mu = e^{\int P dx}$, multiply through, collapse the left side as a product rule, and integrate.

Problem 2.1 — Basic Linear ODE

PROBLEM:

Solve $dy/dx + 2y = e^x$.

SOLUTION:

Step 1 – Standard form: $y' + 2y = e^x$. $P(x) = 2$, $Q(x) = e^x$.

Step 2 – Integrating factor: $\mu = e^{\int 2 dx} = e^{2x}$

Step 3 – Multiply: $e^{2x}y' + 2e^{2x}y = e^{3x} \rightarrow d/dx[e^{2x}y] = e^{3x}$

Step 4 – Integrate: $e^{2x}y = (1/3)e^{3x} + C$

$$y(x) = (1/3)e^x + Ce^{-2x}$$

Problem 2.2 — IVP

PROBLEM:

Solve $x(dy/dx) + 2y = 4x^2$, $y(1) = 3$.

SOLUTION:

Step 1 – Standard form (divide by x): $y' + (2/x)y = 4x$. $P(x) = 2/x$, $Q(x) = 4x$.

Step 2 – $\mu = e^{\int (2/x) dx} = e^{2\ln|x|} = x^2$

Step 3 – Multiply: $x^2y' + 2xy = 4x^3 \rightarrow d/dx[x^2y] = 4x^3$

Step 4 – Integrate: $x^2y = x^4 + C \rightarrow y = x^2 + C/x^2$

Step 5 – Apply $y(1) = 3$: $3 = 1 + C \rightarrow C = 2$

$$y(x) = x^2 + 2/x^2$$

Problem 2.3 — Mixing Problem

PROBLEM:

A 100-L tank initially contains pure water. Brine with 2 kg/L salt flows in at 3 L/min, and the mixed solution drains at 3 L/min. Find the amount of salt $S(t)$.

SOLUTION:

Step 1 – Model: $dS/dt = (\text{rate in}) - (\text{rate out}) = (2 \text{ kg/L})(3 \text{ L/min}) - (S/100)(3 \text{ L/min})$

Step 2 – Simplify: $dS/dt + (3/100)S = 6$. Standard form with $P(t) = 3/100$, $Q(t) = 6$.

Step 3 – $\mu = e^{\int (3/100) dt} = e^{0.03t}$

Step 4 – Multiply and integrate: $d/dt[S \cdot e^{0.03t}] = 6e^{0.03t}$

Step 5 – Integrate: $S \cdot e^{0.03t} = 200e^{0.03t} + C$

Step 6 – Apply $S(0) = 0$: $0 = 200 + C \rightarrow C = -200$

$$S(t) = 200(1 - e^{-0.03t}) \text{ kg}$$

Problem 2.4 — RC Circuit

PROBLEM:

An RC circuit with $R = 1000 \Omega$, $C = 0.001 \text{ F}$ is driven by $V(t) = 12 \text{ V}$ starting at $t = 0$. The capacitor starts uncharged. Find $Q(t)$.

SOLUTION:

Step 1 – Model: $R(dQ/dt) + Q/C = V \rightarrow 1000 Q' + 1000 Q = 12$

Step 2 – Standard form: $Q' + Q = 0.012$. $P = 1$, $Q_{\text{const}} = 0.012$.

Step 3 – $\mu = e^t$

Step 4 – Multiply: $d/dt[Qe^t] = 0.012e^t \rightarrow Qe^t = 0.012e^t + C$

Step 5 – Apply $Q(0) = 0$: $0 = 0.012 + C \rightarrow C = -0.012$

$$Q(t) = 0.012(1 - e^{-t}) \text{ C (steady state: } 0.012 \text{ C} = 12 \text{ mC)}$$

Problem 2.5 — Bernoulli Equation

PROBLEM:

Solve $dy/dx + y = xy^3$. (This is a Bernoulli equation — use the substitution $u = y^{1-n} = y^{-2}$.)

SOLUTION:

Step 1 – Divide by y^3 : $y^{-3}y' + y^{-2} = x$

Step 2 – Substitute $u = y^{-2}$, so $du/dx = -2y^{-3} dy/dx$:

$$(-1/2) du/dx + u = x \rightarrow du/dx - 2u = -2x$$

Step 3 – This is linear in u . $P(x) = -2$, $Q(x) = -2x$.

Step 4 – $\mu = e^{\int (-2) dx} = e^{-2x}$

Step 5 – Multiply: $d/dx[u \cdot e^{-2x}] = -2xe^{-2x}$

Step 6 – Integrate by parts: $u \cdot e^{-2x} = (x + 1/2)e^{-2x} + C$

Step 7 – Back-substitute $u = 1/y^2$: $1/y^2 = x + 1/2 + Ce^{2x}$

$$y(x) = \pm 1/\sqrt{(x + 1/2 + ce^{2x})}$$

SECTION 3

Exact Equations

Write the ODE as $M dx + N dy = 0$. Test for exactness: $\partial M / \partial y = \partial N / \partial x$. If exact, find the potential F by integrating M in x , then match F_y to N .

Problem 3.1 — Basic Exact Equation

PROBLEM:

Solve $(2xy + 3) dx + (x^2 - 1) dy = 0$.

SOLUTION:

Step 1 – Test: $M = 2xy+3$, $N = x^2-1$. $M_y = 2x$, $N_x = 2x$. Exact! ✓

Step 2 – Integrate M in x : $F = \int (2xy+3) dx = x^2y + 3x + g(y)$

Step 3 – Match F_y to N : $F_y = x^2 + g'(y) = x^2 - 1 \rightarrow g'(y) = -1 \rightarrow g(y) = -y$

$$x^2y + 3x - y = C$$

Problem 3.2 — Exact with Initial Condition

PROBLEM:

Solve $(e^x \sin y) dx + (e^x \cos y + y) dy = 0$ with $y(0) = \pi/2$.

SOLUTION:

Step 1 – Test: $M = e^x \sin y$, $N = e^x \cos y + y$. $M_y = e^x \cos y$, $N_x = e^x \cos y$. Exact! ✓

Step 2 – $F = \int (e^x \sin y) dx = e^x \sin y + g(y)$

Step 3 – $F_y = e^x \cos y + g'(y) = e^x \cos y + y \rightarrow g'(y) = y \rightarrow g(y) = y^2/2$

Step 4 – Solution: $e^x \sin y + y^2/2 = C$

Step 5 – Apply $y(0) = \pi/2$: $e^0 \sin(\pi/2) + (\pi/2)^2/2 = C \rightarrow 1 + \pi^2/8 = C$

$$e^x \sin y + y^2/2 = 1 + \pi^2/8$$

Problem 3.3 — Potential from a Conservative Field

PROBLEM:

Find the potential function $F(x,y)$ for the conservative vector field $F = (2xy + y^2, x^2 + 2xy)$. Verify it is conservative.

SOLUTION:

Step 1 – $M = 2xy + y^2$, $N = x^2 + 2xy$. $M_y = 2x + 2y$, $N_x = 2x + 2y$.
Conservative! ✓

Step 2 – $F = \int (2xy + y^2) dx = x^2y + xy^2 + g(y)$

Step 3 – $F_y = x^2 + 2xy + g'(y) = x^2 + 2xy \rightarrow g'(y) = 0 \rightarrow g(y) = C_0$

$$F(x, y) = x^2y + xy^2 + C_0$$

Problem 3.4 — Microeconomics: Cobb-Douglas Indifference

PROBLEM:

A consumer has utility $U(x,y) = x^2y$. Derive the exact ODE for the indifference curves and find the family of curves.

SOLUTION:

Step 1 – Marginal utilities: $MU_x = 2xy$, $MU_y = x^2$

Step 2 – Total differential: $dU = 2xy dx + x^2 dy = 0$ along indifference curves

Step 3 – Verify exactness: $M_y = 2x = N_x$ ✓

Step 4 – $F = \int 2xy dx = x^2y + g(y)$. $F_y = x^2 + g'(y) = x^2 \rightarrow g'(y) = 0$.

$$x^2y = C \text{ (family of indifference curves)}$$

Problem 3.5 — Thermodynamics: Exact Heat Differential

PROBLEM:

For an ideal gas, $dQ = C_V dT + (R T/V) dV$. Verify this is exact and find the potential (entropy S).

SOLUTION:

Step 1 – $M = C_V$, $N = RT/V$. $M_V = 0$, $N_T = R/V$. Not exact as written!

Step 2 – Divide by T (this is the integrating factor $1/T$):

$$(C_V/T) dT + (R/V) dV = 0$$

Step 3 – Now $M_V = 0$, $N_T = 0$. Exact! ✓

Step 4 – $F = \int (C_V/T) dT = C_V \ln T + g(V)$

Step 5 – $F_V = g'(V) = R/V \rightarrow g(V) = R \ln V$

$$S(T, V) = C_V \ln T + R \ln V + S_0 \text{ (entropy)}$$

Non-Exact Equations & Integrating Factors

When $M_y \neq N_x$, test for an integrating factor: if $(M_y - N_x)/N$ is a function of x alone, use $\mu(x) = e^{\int f(x) dx}$. If $(N_x - M_y)/M$ is a function of y alone, use $\mu(y) = e^{\int g(y) dy}$.

Problem 4.1 — x-only Integrating Factor

PROBLEM:

Solve $(2x^2 - y) dx + (x^2y + x) dy = 0$.

SOLUTION:

Step 1 – Test: $M = 2x^2 - y$, $N = x^2y + x$. $M_y = -1$, $N_x = 2xy + 1$. Not exact.

Step 2 – Test x-only IF: $(M_y - N_x)/N = (-1 - 2xy - 1)/(x^2y + x) = -2(1+xy)/(x(xy+1)) = -2/x$

Step 3 – $\mu(x) = e^{\int (-2/x) dx} = e^{-2\ln|x|} = 1/x^2$

Step 4 – Multiply: $(2 - y/x^2) dx + (y + 1/x) dy = 0$. Verify: $M^*_y = -1/x^2 = N^*_x$ ✓

Step 5 – $F = \int (2 - y/x^2) dx = 2x + y/x + g(y)$

Step 6 – $F_y = 1/x + g'(y) = y + 1/x \rightarrow g'(y) = y \rightarrow g(y) = y^2/2$

$$2x + y/x + y^2/2 = C$$

Problem 4.2 — y-only Integrating Factor

PROBLEM:

Solve $(2x^2y) dx + (x^3 + y^3) dy = 0$.

SOLUTION:

Step 1 – Test: $M = 2x^2y$, $N = x^3 + y^3$. $M_y = 2x^2$, $N_x = 3x^2$. Not exact.

Step 2 – Test x-only IF: $(M_y - N_x)/N = (2x^2 - 3x^2)/(x^3 + y^3) = -x^2/(x^3 + y^3)$. Depends on $y \rightarrow$ no.

Step 3 – Test y-only IF: $(N_x - M_y)/M = (3x^2 - 2x^2)/(2x^2y) = x^2/(2x^2y) = 1/(2y)$

Step 4 – $\mu(y) = e^{\int (1/2y) dy} = e^{(1/2)\ln|y|} = \sqrt{|y|} = y^{1/2}$

Step 5 – Multiply: $2x^2y^{3/2} dx + (x^3y^{1/2} + y^{7/2}) dy = 0$

Step 6 – Verify: $M^*_y = 3x^2y^{1/2} = N^*_x$ ✓

Step 7 – $F = (2/3)x^3y^{3/2} + g(y)$. $F_y = x^3y^{1/2} + g'(y) = x^3y^{1/2} + y^{7/2}$

Step 8 – $g'(y) = y^{7/2} \rightarrow g(y) = (2/9)y^{9/2}$

$$\left(\frac{2}{3}\right)x^3y^{3/2} + \left(\frac{2}{9}\right)y^{9/2} = C$$

Problem 4.3 — Homogeneous Equation via IF

PROBLEM:

Solve $(x^2 + y^2) dx + (x^2 - xy) dy = 0$.

SOLUTION:

Step 1 – Test: $M_y = 2y$, $N_x = 2x - y$. Not exact.

Step 2 – Test x-only: $(M_y - N_x)/N = (2y - 2x + y)/(x^2 - xy) = (3y - 2x)/(x(x-y))$. Not x-only.

Step 3 – Test y-only: $(N_x - M_y)/M = (2x - y - 2y)/(x^2 + y^2) = (2x - 3y)/(x^2 + y^2)$. Not y-only.

Step 4 – This is a homogeneous equation (both M and N are degree 2). Use substitution $y = vx$, $dy = v dx + x dv$:

$$(x^2 + v^2x^2) dx + (x^2 - vx^2)(v dx + x dv) = 0$$

Step 5 – Divide by x^2 : $(1 + v^2) dx + (1 - v)(v dx + x dv) = 0$

Step 6 – Collect: $[(1 + v^2) + v(1-v)] dx + x(1-v) dv = 0 \rightarrow (1 + v) dx + x(1-v) dv = 0$

Step 7 – Separate: $(1-v)/(1+v) dv = -dx/x$. Integrate: $v - 2\ln|1+v| = -\ln|x| + C$

Step 8 – Back-substitute $v = y/x$ and simplify:

$$(y/x) - 2\ln|1 + y/x| + \ln|x| = C$$

Problem 4.4 — Chemistry: Rate Law with IF

PROBLEM:

A second-order reaction $A + B \rightarrow P$ has rate $d[A]/dt = -k[A][B]$ with $[A]_0 = a$, $[B]_0 = b$. Express $[A]$ as a function of $[B]$.

SOLUTION:

Step 1 – Let x = amount reacted. Then $[A] = a - x$, $[B] = b - x$, and $dx/dt = k(a-x)(b-x)$.

Step 2 – As a differential form: $k(a-x)(b-x) dt - dx = 0$. Rewrite as: $dx - k(a-x)(b-x) dt = 0 \rightarrow M = 1$, $N = -k(a-x)(b-x)$. (Treat x as dependent, t as independent.)

Step 3 – Better approach: treat as separation. $dx/[(a-x)(b-x)] = k dt$

Step 4 – Partial fractions: $1/[(a-x)(b-x)] = (1/(b-a))[1/(a-x) - 1/(b-x)]$

Step 5 – Integrate: $(1/(b-a)) [\ln|(a-x)/(b-x)|] = kt + C_0$

Step 6 – Apply $x(0) = 0$: $(1/(b-a)) \ln(a/b) = C_0$

$$\ln[(a-x)b / ((b-x)a)] = (b-a)kt \rightarrow [A] = a - x \text{ where} \\ (a-x)/(b-x) = (a/b)e^{(b-a)kt}$$

Problem 4.5 — Engineering: Heat with IF

PROBLEM:

Solve $(e^{x/y} + x e^{x/y}/y) dx - (x^2 e^{x/y}/y^2) dy = 0$.

SOLUTION:

Step 1 – Factor: $e^{x/y}[(1 + x/y) dx - (x^2/y^2) dy] = 0$

Step 2 – Test exactness: $M = e^{x/y}(1 + x/y)$, $N = -x^2 e^{x/y}/y^2$

Step 3 – $M_y = e^{x/y}[-x/y^2 + (-1/y^2 + x/y^3) \cdot y] \dots$ (complex). Use the quotient test:

After computation: $(N_x - M_y)/M = 1/y$ (a function of y alone!).

Step 4 – $\mu(y) = e^{\int (1/y) dy} = y$

Step 5 – Multiply by y : $ye^{x/y}(1 + x/y) dx - (x^2 e^{x/y}/y) dy = 0$

Step 6 – Simplify: $(y + x)e^{x/y} dx - (x^2/y)e^{x/y} dy = 0$. Verify exact.

Step 7 – $F = \int (y+x)e^{x/y} dx$. With $u = x/y$: $F = y^2 e^{x/y} + g(y)$.

Step 8 – $F_y = 2ye^{x/y} - xe^{x/y} + g'(y) = N \rightarrow g(y) = 0$

$$y^2 e^{x/y} = C$$

Numerical Methods (Euler, Heun, RK4)

For ODEs that cannot be solved analytically, step along the slope field. Euler uses 1 slope eval per step ($O(h)$ global error), Heun uses 2 ($O(h^2)$ global), RK4 uses 4 ($O(h^4)$ global).

Problem 5.1 — Euler's Method by Hand

PROBLEM:

Use Euler's method with $h = 0.1$ to approximate $y(0.3)$ for $dy/dx = x + y$, $y(0) = 1$. Compare to the exact solution $y = 2e^x - x - 1$.

SOLUTION:

$$\text{Euler formula: } y_{n+1} = y_n + h \cdot f(x_n, y_n) = y_n + 0.1 \cdot (x_n + y_n)$$

$$\text{Step 1 — } x_0=0, y_0=1: y_1 = 1 + 0.1(0+1) = 1.1000$$

$$\text{Step 2 — } x_1=0.1, y_1=1.1: y_2 = 1.1 + 0.1(0.1+1.1) = 1.2200$$

$$\text{Step 3 — } x_2=0.2, y_2=1.22: y_3 = 1.22 + 0.1(0.2+1.22) = 1.3620$$

$$\text{Step 4 — Exact: } y(0.3) = 2e^{0.3} - 0.3 - 1 \approx 2(1.3499) - 1.3 \approx 1.3997$$

$$\text{Step 5 — Error: } |1.3620 - 1.3997| = 0.0377 \text{ (2.7\% error)}$$

$$\text{Euler: } y(0.3) \approx 1.3620 \mid \text{Exact: } 1.3997 \mid \text{Error: } 2.7\%$$

Problem 5.2 — Heun's Method

PROBLEM:

Repeat Problem 5.1 using Heun's method (Improved Euler) with $h = 0.1$.

SOLUTION:

$$\text{Predictor: } \tilde{y}_{n+1} = y_n + h \cdot f(x_n, y_n)$$

$$\text{Corrector: } y_{n+1} = y_n + (h/2) \cdot [f(x_n, y_n) + f(x_{n+1}, \tilde{y}_{n+1})]$$

$$\text{Step 1 — Predict: } \tilde{y}_1 = 1 + 0.1(0+1) = 1.1$$

$$\text{Step 2 — Correct: } y_1 = 1 + 0.05 \cdot [(0+1) + (0.1+1.1)] = 1 + 0.05 \cdot 2.2 = 1.1100$$

$$\text{Step 3 — Predict: } \tilde{y}_2 = 1.11 + 0.1(0.1+1.11) = 1.2310$$

$$\text{Step 4 — Correct: } y_2 = 1.11 + 0.05 \cdot [(0.1+1.11) + (0.2+1.231)] = 1.11 + 0.05 \cdot 2.641 = 1.2421$$

$$\text{Step 5 — Predict: } \tilde{y}_3 = 1.2421 + 0.1(0.2+1.2421) = 1.3863$$

$$\text{Step 6 — Correct: } y_3 = 1.2421 + 0.05 \cdot [(0.2+1.2421) + (0.3+1.3863)] = 1.2421 + 0.05 \cdot 3.1284 = 1.3985$$

Step 7 – Error: $|1.3985 - 1.3997| = 0.0012$ (0.09% error – 31x better than Euler!)

Heun: $y(0.3) \approx 1.3985$ | Exact: 1.3997 | Error: 0.09%

Problem 5.3 — RK4 by Hand (One Step)

PROBLEM:

Compute ONE step of RK4 from (0, 1) with $h = 0.1$ for $dy/dx = x + y$. Compare the result to the exact $y(0.1) = 2e^{0.1} - 0.1 - 1$.

SOLUTION:

$$k_1 = f(0, 1) = 0 + 1 = 1.0000$$

$$k_2 = f(0.05, 1 + 0.05 \cdot 1) = f(0.05, 1.05) = 0.05 + 1.05 = 1.1000$$

$$k_3 = f(0.05, 1 + 0.05 \cdot 1.1) = f(0.05, 1.055) = 0.05 + 1.055 = 1.1050$$

$$k_4 = f(0.1, 1 + 0.1 \cdot 1.105) = f(0.1, 1.1105) = 0.1 + 1.1105 = 1.2105$$

$$y_1 = 1 + (0.1/6)(1 + 2 \cdot 1.1 + 2 \cdot 1.105 + 1.2105)$$

$$y_1 = 1 + (0.01667)(6.6205) = 1 + 0.11034 = 1.11034$$

$$\text{Exact: } y(0.1) = 2e^{0.1} - 1.1 \approx 2(1.10517) - 1.1 = 1.11034$$

RK4: $y(0.1) \approx 1.11034$ | Exact: 1.11034 | Error $< 10^{-6}$

Problem 5.4 — Unsolved Numerically: $y' = \sin(xy)$

PROBLEM:

The ODE $dy/dx = \sin(xy)$ has no closed-form solution. Use RK4 with $h = 0.2$ to approximate $y(0.6)$ given $y(0) = 1$. Set up the computation (you need not compute all values by hand — describe what a computer would do).

SOLUTION:

Step 1 – $f(x, y) = \sin(xy)$. At each step, compute:

$$k_1 = \sin(x_n \cdot y_n)$$

$$k_2 = \sin((x_n + h/2)(y_n + h \cdot k_1/2))$$

$$k_3 = \sin((x_n + h/2)(y_n + h \cdot k_2/2))$$

$$k_4 = \sin((x_n + h)(y_n + h \cdot k_3))$$

$$y_{n+1} = y_n + (h/6)(k_1 + 2k_2 + 2k_3 + k_4)$$

Step 2 – Step $0 \rightarrow 0.2$: $k_1 = \sin(0) = 0$, $k_2 = \sin(0.1 \cdot 1) = 0.0998$,
 $k_3 = \sin(0.1 \cdot 1.01) = 0.1008$, $k_4 = \sin(0.2 \cdot 1.020) = 0.1997$

$$y_1 = 1 + (0.2/6)(0 + 0.1997 + 0.2016 + 0.1997) \approx 1 + 0.02670 \approx 1.0267$$

Step 3 – Continuing with a computer for steps 2 and 3 ($h=0.2$, 3 steps total to reach $x=0.6$):

$$y_2 \approx 1.0537 \text{ (at } x=0.4), y_3 \approx 1.0808 \text{ (at } x=0.6)$$

$$\text{RK4 (h=0.2): } y(0.6) \approx 1.0808$$

Problem 5.5 — Error Comparison & Step-Size Choice

PROBLEM:

For $dy/dx = y$, $y(0) = 1$, the exact solution is $y = e^x$. Using $h = 0.1$ on $[0, 1]$, rank Euler, Heun, and RK4 by absolute error at $x = 1$. How small must h be for Euler to match RK4's accuracy with $h = 0.1$?

SOLUTION:

$$\text{Exact: } y(1) = e \approx 2.71828$$

$$\text{Euler (h=0.1, 10 steps): } y_{10} = (1.1)^{10} \approx 2.5937. \text{ Error} = 0.1246$$

$$\text{Heun (h=0.1): } y_{10} \approx 2.7141. \text{ Error} = 0.0042$$

$$\text{RK4 (h=0.1): } y_{10} \approx 2.71828. \text{ Error} < 10^{-7}$$

$$\text{Ranking: RK4 (best) > Heun > Euler}$$

For Euler to achieve $\text{Error} \sim 10^{-7}$, since $\text{global error} = O(h)$, we need $h \approx 10^{-7}$.

That means ~10 million steps – vs RK4's 10 steps. RK4 is ~1 million times more efficient here.

Euler err=0.1246 | Heun err=0.0042 | RK4 err<10⁻⁷. Euler needs $h \approx 10^{-7}$ to match RK4 at $h=0.1$.